Middle Miocene magnetostratigraphy of the Pebas Formation at Tamshiyacu and Santa Carina (Iquitos, Peru): a C5AAr (∼13.2–13.4 Ma) tie for the Pan-Amazonian wetland and its *Purussaurus* record

N. Anderson and co-authors
[affiliations]
Corresponding author: nickstephann@gmail.com

Abstract

The middle Miocene Pebas Formation records the Pan-Amazonian "mega-wetland," and its lignite beds yield abundant vertebrate fossils — yet the timing and tempo of that record remain poorly constrained, with age uncertainties of a million years or more, because the strata are rarely tied directly to the geomagnetic polarity time scale (GPTS). We present paleomagnetic u-channel records from two cutbank sections ∼170 km apart near Iquitos, Peru — Tamshiyacu (TAM) and Santa Carina (SC), the localities of Gingras et al. (2002) — each preserving a single, well-resolved polarity reversal. Reconstruction of core azimuth recovers geographic declination, so the reversal is expressed antipodally in both inclination and declination. Tied by the reversal, both sections share a normal-to-reversed transition — the C5ABn–C5AAr boundary — whereas their lignite flooding surfaces do not align and are demonstrably diachronous. Four independent constraints — the polarity pattern, a palynological floor (>12.7 Ma), a *Purussaurus* ceiling (<13.8 Ma), and U-Pb detrital-zircon ages from the *Purussaurus neivensis*–bearing La Venta section (youngest zircon populations ∼13.9 Ma; Mora-Rojas et al., 2023) — converge on **chron C5AAr, ∼13.2–13.4 Ma**. A *Purussaurus* cranial horizon discovered at Tamshiyacu in 2025 falls within C5AAr, dating it to a ∼180 kyr window and to the same chron as the species at La Venta — a sub-million-year calibration relevant to evolutionary and biogeographic models. Down-core X-ray fluorescence shows iron is dominantly authigenic, recording reducing, episodically brackish bottom water; net accumulation is very low (∼3 m/Myr), marking a condensed, incomplete succession. These records provide the first direct GPTS calibration of the Pebas at Iquitos and a chronologic bridge to the La Venta fauna.

INTRODUCTION

The Miocene reorganization of northern South America — Andean uplift, the Pebas mega-wetland, and the eventual birth of the transcontinental Amazon — is central to Neotropical biogeography (Hoorn et al., 2010a; Wesselingh & Salo, 2006). The formation's lignites and shell beds yield an abundant vertebrate fauna — crocodylians, turtles, rodents, and the giant caiman *Purussaurus* — recovered almost yearly from actively eroding cutbanks; the Iquitos Pebas alone has produced a hyperdiverse crocodylian community, including the ∼13 Ma gavialoid *Gryposuchus pachakamue* whose skull records parallel evolution in the proto-Amazonian mega-wetland (Salas-Gismondi et al., 2015, 2016). Yet individual fossil horizons typically carry age uncertainties of a million years or more, because the age framework of the Pebas Formation rests largely on molluscan and palynological biostratigraphy and sparse radiometric ages and its strata are seldom tied directly to the GPTS. Such coarse control limits the fauna's use in the evolutionary and biogeographic models that increasingly demand sub-million-year resolution to test rates of diversification, dispersal, and body-size evolution. Whether Pebas deposition was lacustrine and essentially freshwater (Wesselingh & Salo, 2006) or tidally influenced and periodically brackish-marine (Räsänen et al., 1995; Gingras et al., 2002) has been debated for decades, and the correlation of scattered outcrops to one another and to dated faunas such as La Venta remains loose.

Here we calibrate two Pebas sections near Iquitos, Peru, to the GPTS using paleomagnetic u-channel data, and integrate the result with sedimentology, ichnology (after Gingras et al., 2002), down-core geochemistry, and vertebrate and U-Pb age control. The sections are the same cutbanks — Tamshiyacu and Santa Julia II (here "Santa Carina") — whose ichnology first argued for tidal influence (Gingras et al., 2002), allowing our magnetostratigraphy to be read directly against an established facies and process model.

GEOLOGIC SETTING AND METHODS

The Pebas Formation (≈Solimões Fm. of Brazil) accumulated in the Andean foreland/back-arc of western Amazonia during the middle Miocene, as a low-gradient system of shallow lakes, swamps, and floodplains periodically connected to the sea (Hoorn, 1993; Hoorn et al., 2010a,b). At Tamshiyacu ($4.01°$S, $73.15°$W) and Santa Carina ($3.36°$S, $71.79°$W; Fig. 1) the formation comprises blue-grey laminated and massive muds (facies F1A/F1B of Gingras et al., 2002), heterolithic silty sands (F2A–F2D), rooted lignites (F3A), winnowed mollusc shell beds (F3B), and *Glossifungites*-demarcated firmgrounds (F3C), organized into lignite-capped parasequences.

Oriented u-channels were sampled from continuous mud-dominated intervals and measured at Oregon State University. Characteristic remanent magnetization (ChRM) was isolated from despiked, high-coercivity (30–100 mT) steps by principal-component analysis; at $\sim 4°$S the geocentric axial dipole field is shallow, so normal polarity corresponds to negative inclination. Because cores were azimuthally unoriented in the laboratory, we reconstructed geographic declination from field measurements of each core's outward face plus a single fixed reference rotation, validated by the collapse of virtual geomagnetic poles (VGPs) toward the spin axis. Bed-by-bed lithologic logs were rendered with `pystrat`. Down-core geochemistry was measured by energy-dispersive X-ray fluorescence (XRF) core scanning.

RESULTS

Polarity records. Each section preserves shallow, dual-polarity ChRM directions and a single robust reversal. Tamshiyacu is normal-below, reversed-above; Santa Carina is normal–reversed–normal. Azimuth reconstruction expresses the reversal antipodally in both inclination and declination and concentrates the VGPs near the poles, confirming a primary detrital remanence. At Tamshiyacu the uppermost core group (the *Purussaurus*-bearing lignite and capping beds) is NRM-only with poor susceptibility and yields no interpretable direction; usable polarity is therefore restricted to the lower $\sim$2.5 m (Fig. 2).

Shared reversal. Hung on the reversal, both sections show the same normal-to-reversed transition at the datum — the C5ABn–C5AAr boundary — with Santa Carina continuing up into the succeeding normal chron (C5AAn) and Tamshiyacu truncated within C5AAr (Fig. 2). The lignite flooding surfaces do not coincide once the reversals are matched, demonstrating that they are diachronous facies boundaries.

Facies gradient. Tamshiyacu is mud- and silt-dominated (86% of the section; no clean sand, no concretions), whereas Santa Carina is coarser and more fossiliferous (a metre-scale sand body, additional shell beds, and carbonate concretions), consistent with a more basinward bay-centre and a more marginal bay-margin position, respectively, on a low-gradient shoreline.

Geochemistry. XRF shows that total Fe does not track magnetic-mineral concentration (magnetic susceptibility) but instead co-varies with S and Ca+Sr: iron is dominantly authigenic, held in pyrite and carbonate (Fig. 3). Discrete S–Fe–Ca-enriched intervals mark reducing, shell-bearing horizons; intervening beds are more oxidized and detritally-diluted.

DISCUSSION

Age. Four independent constraints converge on chron C5AAr, ∼13.2–13.4 Ma: (i) the shared reversal and the N–R–N thickness pattern; (ii) a palynological floor (zones T-15/T-16, >12.7 Ma; Jaramillo et al., 2011); (iii) a *Purussaurus* maximum-age ceiling (<13.8 Ma); and (iv), most decisively, U-Pb detrital-zircon geochronology of the *Purussaurus neivensis*–bearing La Venta section (Honda Group), whose fossil-rich upper La Victoria Formation yields youngest zircon populations of ∼13.9 Ma (sample 44011, YSP 13.9±0.3 Ma) and whose C5AAr interval there spans 13.18–13.36 Ma (Mora-Rojas et al., 2023) — an anchor external to the Pebas. The Tamshiyacu *Purussaurus* cranial horizon falls within C5AAr — the same chron in which the species occurs at La Venta — linking the Iquitos Pebas directly to the classic Laventan vertebrate fauna. The tie is independently consistent with molluscan biostratigraphy: the hyperdiverse Iquitos crocodylian assemblage, including *Gryposuchus pachakamue*, is assigned to Mollusc Zone 8 (∼13 Ma) in the same sections (Salas-Gismondi et al., 2016), and our C5AAr calibration now places that assemblage on the GPTS to ∼180 kyr.

Environment. The authigenic Fe–S–Ca geochemistry records a reducing, sporadically dysaerobic bottom water in which bacterial sulphate reduction pyritized the mud; because sulphate is scarce in freshwater, the punctuated pyrite enrichments most plausibly track brief brackish incursions, and the Ca+Sr shell beds are their winnowed transgressive lags. This independently supports the tidally-influenced, brackish bay-margin interpretation from ichnology and tidal rhythmites at the same localities (Gingras et al., 2002; Räsänen et al., 1995), superimposed on a dominantly freshwater mega-wetland (Hoorn et al., 2010a; Wesselingh & Salo, 2006).

Condensation and correlation. Net accumulation is very low: the C5AAr chron (∼180 kyr) is expressed in only ∼0.5 m at Santa Carina (∼3 m/Myr compacted; ∼6–9 m/Myr decompacted). This is an order of magnitude below the ∼100 m/Myr implied by orbital (obliquity) cyclicity elsewhere in the Pebas (Hoorn et al., 2022) and the ∼45–130 m/Myr regional average from the formation's ∼350–1000 m thickness (Hoorn et al., 2010a). The discrepancy is diagnostic of *incompleteness*: at a sediment-starved, low-accommodation bay margin (Gingras et al., 2002), rapid intra-cycle deposition was repeatedly interrupted by non-deposition, transgressive erosion (the *Glossifungites* firmgrounds), and winnowing (shell lags), so net accumulation is slow. The same low-accommodation architecture makes lignite flooding surfaces diachronous and explains why the two sections correlate only loosely by lithology; the reversal provides the isochronous tie (Fig. 2), and its offset from the lignite ties quantifies the diachroneity. A regional reversal "barcode" comparison to La Venta reinforces the C5AAr correlation (Fig. 4).

Two independent checks support the calibration. First, the low accumulation rate is not anomalous: the shipboard age model for ODP Site 1237 (Nazca Ridge; Mix et al., 2003) gives ∼6–9 m/Myr through 12–15 Ma (∼7.7 m/Myr at 13 Ma), essentially our decompacted value, so a condensed Pebas margin at ∼13 Ma is expected. Second, ODP 1237 records the C5AAn–C5AAr normal-to-reversed transition at ∼13.2–13.3 Ma (Fig. 5), independently confirming the age of the reversal we tie to.

CONCLUSIONS

Paleomagnetic records from Tamshiyacu and Santa Carina provide the first direct GPTS calibration of the Pebas Formation at Iquitos. Both sections share the C5ABn–C5AAr reversal, and four independent constraints place the studied interval in chron C5AAr, ∼13.2–13.4 Ma. A *Purussaurus* cranial horizon at Tamshiyacu lies within C5AAr, tying the Iquitos Pebas to the U-Pb-dated La Venta occurrence of *P. neivensis*. Authigenic Fe–S–Ca geochemistry confirms reducing, episodically brackish conditions consistent with a tidally-influenced bay margin, and very low net accumulation (∼3 m/Myr) identifies the record as condensed and incomplete — the expected signature of a low-accommodation Pan-Amazonian wetland margin. More

broadly, by pinning a single fossil horizon to a ∼180 kyr polarity chron and to the U-Pb-dated La Venta occurrence of *P. neivensis*, the approach converts an abundant but coarsely dated fauna into a sub-million-year chronologic framework — resolution suitable for calibrating Neotropical evolutionary and biogeographic models, and a template for dating the many other fossiliferous Pebas cutbanks that at present lack direct GPTS control.

ACKNOWLEDGMENTS

[Funding; field and laboratory assistance; OSU paleomagnetism facility; *Purussaurus* identification (Antoine, pers. comm.).]

REFERENCES CITED

Gingras, M.K., Räsänen, M.E., Pemberton, S.G., Romero, L.P., 2002, Ichnology and sedimentology reveal depositional characteristics of bay-margin parasequences in the Miocene Amazonian foreland basin: Journal of Sedimentary Research, v. 72, p. 871–883.

Hoorn, C., 1993, Marine incursions and the influence of Andean tectonics on the Miocene depositional history of northwestern Amazonia: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 105, p. 267–309.

Hoorn, C., et al., 2010a, Amazonia through time: Andean uplift, climate change, landscape evolution, and biodiversity: Science, v. 330, p. 927–931.

Hoorn, C., Wesselingh, F.P., Hovikoski, J., Guerrero, J., 2010b, The development of the Amazonian mega-wetland (Miocene; Brazil, Colombia, Peru, Bolivia), *in* Hoorn, C., and Wesselingh, F.P., eds., Amazonia: Landscape and Species Evolution: A Look into the Past: Oxford, Wiley-Blackwell, p. 123–142.

Hoorn, C., Kukla, T., Bogotá-Angel, G., et al., 2022, Cyclic sediment deposition by orbital forcing in the Miocene wetland of western Amazonia? New insights from a multidisciplinary approach: Global and Planetary Change, v. 210, 103717.

Jaramillo, C., Rueda, M., Torres, V., 2011, A palynological zonation for the Cenozoic of the Llanos and Llanos Foothills of Colombia: Palynology, v. 35, p. 46–84.

Mix, A.C., Tiedemann, R., Blum, P., et al., 2003, Proceedings of the Ocean Drilling Program, Initial Reports, v. 202: College Station, Texas, Ocean Drilling Program.

Mora-Rojas, L., Cárdenas, A., Jaramillo, C., et al., 2023, Stratigraphy of a middle Miocene neotropical Lagerstätte (La Venta Site, Colombia): Geodiversitas, v. 45(6), p. 197–221.

Räsänen, M.E., Linna, A.M., Santos, J.C.R., Negri, F.R., 1995, Late Miocene tidal deposits in the Amazonian foreland basin: Science, v. 269, p. 386–390.

Salas-Gismondi, R., Flynn, J.J., Baby, P., Tejada-Lara, J.V., Wesselingh, F.P., Antoine, P.-O., 2015, A Miocene hyperdiverse crocodylian community reveals peculiar trophic dynamics in proto-Amazonian mega-wetlands: Proceedings of the Royal Society B, v. 282, 20142490.

Salas-Gismondi, R., Flynn, J.J., Baby, P., Tejada-Lara, J.V., Claude, J., Antoine, P.-O., 2016, A new 13 million year old gavialoid crocodylian from proto-Amazonian mega-wetlands reveals parallel evolutionary trends in skull shape linked to longirostry: PLoS ONE, v. 11, no. 4, e0152453.

Wesselingh, F.P., Salo, J.A., 2006, A Miocene perspective on the evolution of the Amazonian biota: Scripta Geologica, v. 133, p. 439–458.

FIGURE CAPTIONS

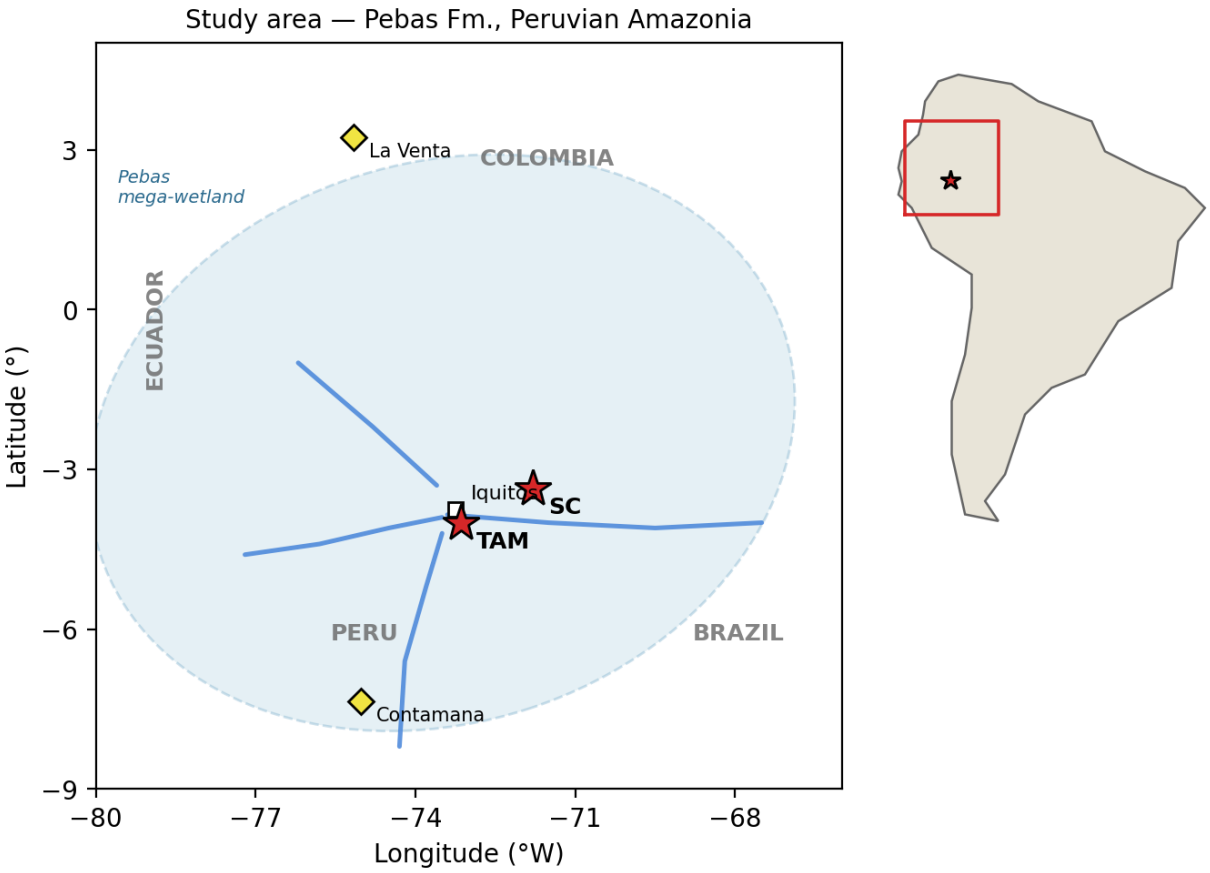
Figure 1. Location of the Tamshiyacu (TAM) and Santa Carina (SC) sections along the Amazon River near Iquitos, Peru, with the regional Miocene localities La Venta and Contamana and the approximate extent of the Pebas mega-wetland. Political boundaries and river courses schematic; site markers at true coordinates.

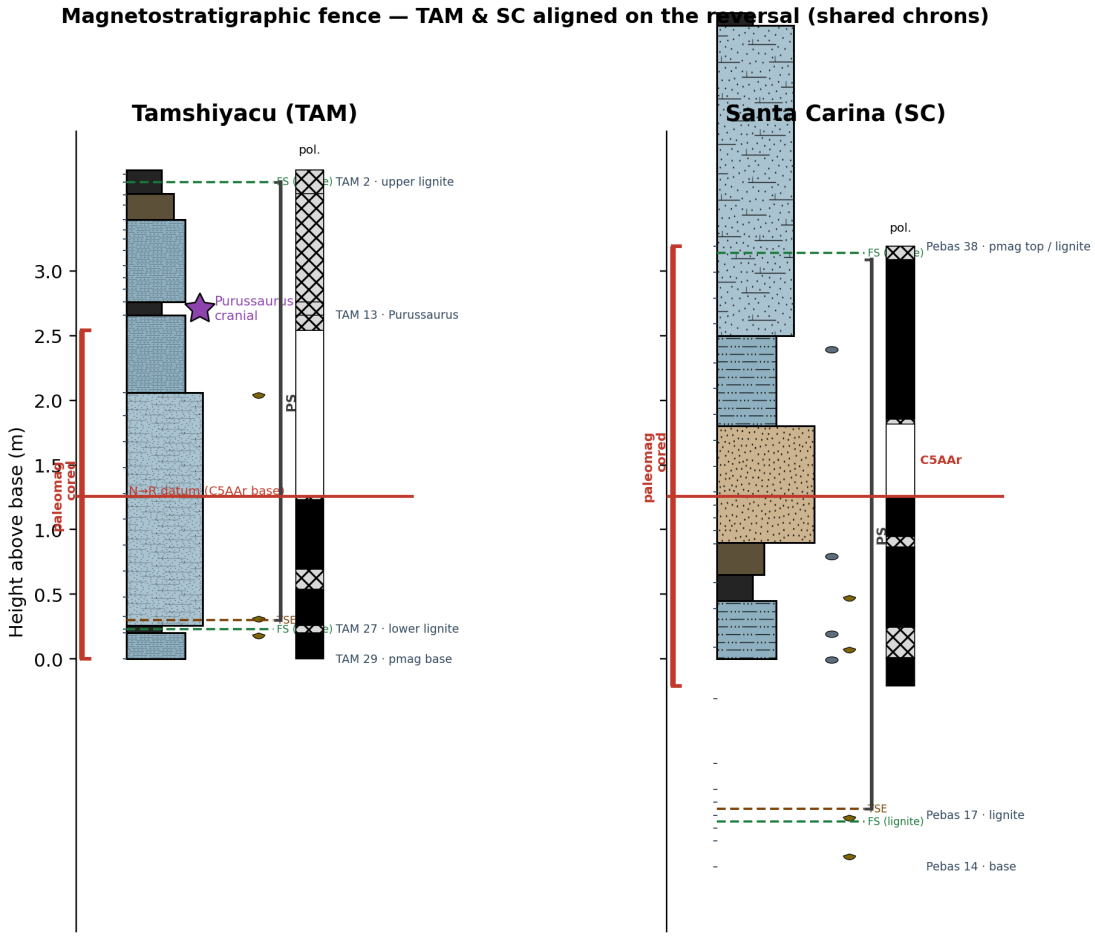
Figure 2. Magnetostratigraphic fence: the TAM and SC lithologic columns (`pystrat`; facies after Gingras et al., 2002) with their polarity records hung on the shared N→R reversal (C5AAr base). Both are normal below and reversed at the datum (shared C5ABn–C5AAr boundary); lignite flooding surfaces are offset (diachronous). Hatching denotes intervals with no reliable remanence (lignites; the NRM-only Group 1 at TAM, which hosts the *Purussaurus* cranial).

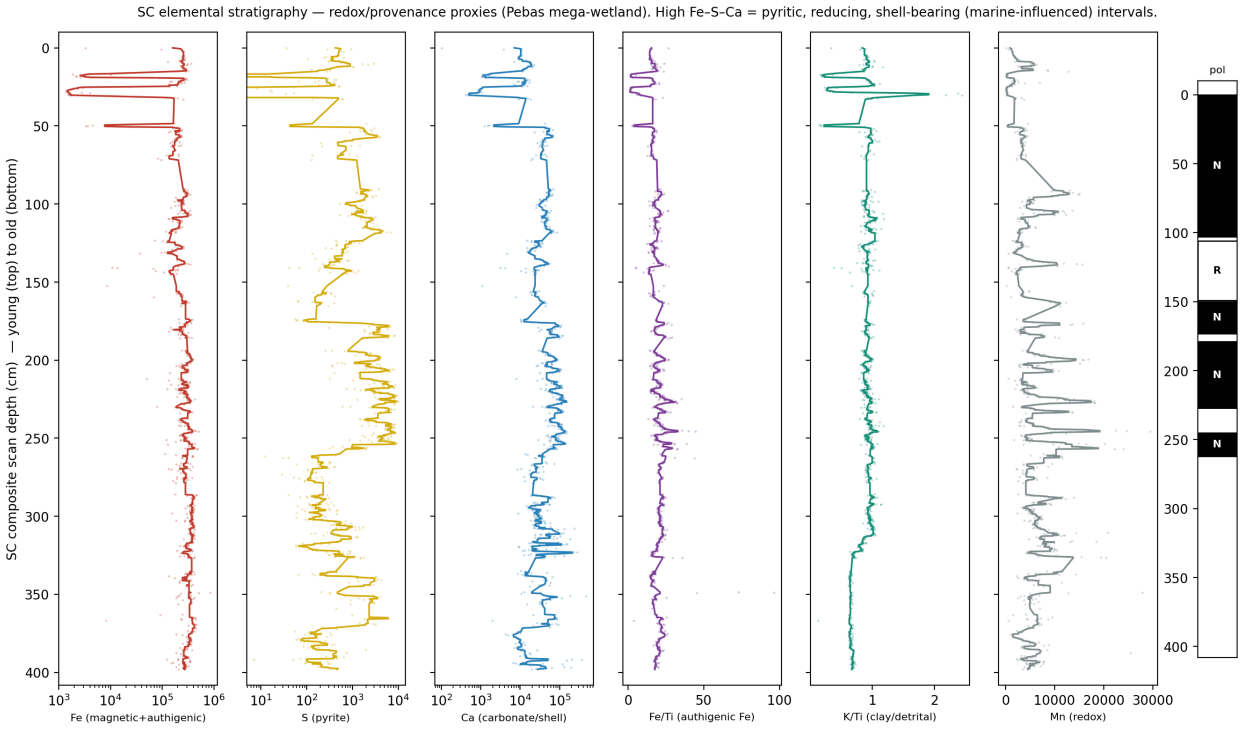
Figure 3. Down-core XRF for the SC composite: Fe, S, Ca (log scale), Fe/Ti, K/Ti, and Mn against depth, with the polarity column. Co-enriched S–Fe–Ca intervals mark reducing, pyritic, shell-bearing (brackish-influenced) horizons; total Fe is authigenic and does not track magnetic concentration.

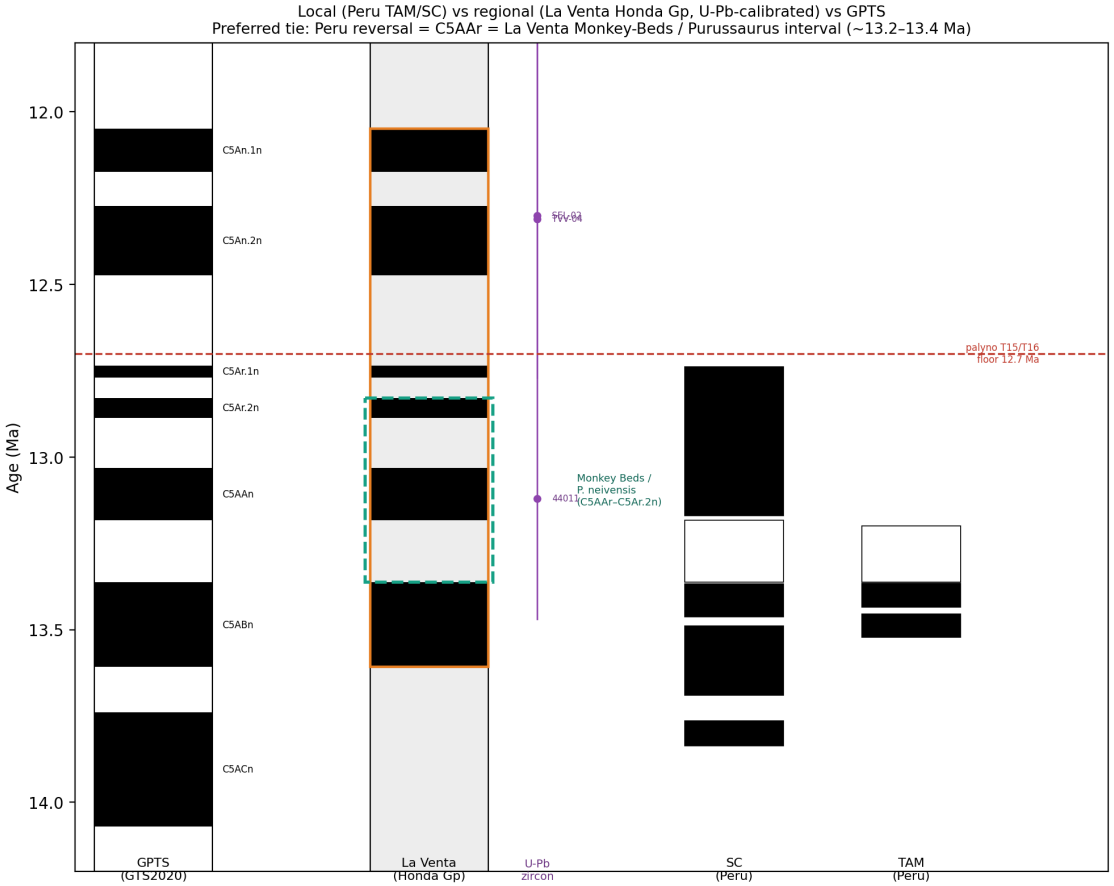
Figure 4. Reversal "barcode" correlation of the Iquitos sections to the GPTS and to the U-Pb-dated La Venta section, supporting a C5AAr (∼13.2–13.4 Ma) tie.

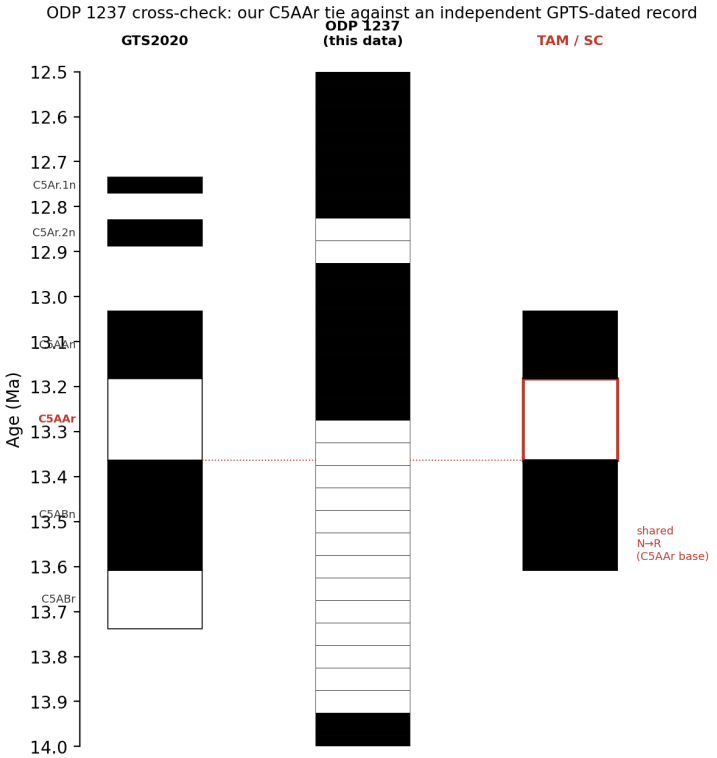
Figure 5. Cross-check against ODP Site 1237 (Leg 202, Nazca Ridge; Mix et al., 2003). GTS2020 chrons (left); polarity derived from the Site 1237 AF-demag data on the shipboard age-depth model (centre); and the interval calibrated by our sections (right). ODP 1237 independently records the C5AAn–C5AAr transition at ∼13.2–13.3 Ma and yields a mid-Miocene accumulation rate (∼7.7 m/Myr at 13 Ma) matching our decompacted estimate. The Site 1237 column is limited by the 1-Myr age-model ties and does not resolve the thin C5ABn normal. (May be placed as a supplementary figure.)


Study area — Pebas Fm., Peruvian Amazonia
La Venta
COLOMBIA
Pebas
mega-wetland
ECUADOR
Iquitos
SC
TAM
PERU
BRAZIL
Contamana
Latitude (°)
Longitude (°W)
3
0
−3
−6
−9
−80
−77
−74
−71
−68


Magnetostratigraphic fence — TAM & SC aligned on the reversal (shared chrons)
Tamshiyacu (TAM)
Santa Carina (SC)
Height above base (m)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
pol.
FS
TAM 2 · upper lignite
Purussaurus cranial
TAM 13 · Purussaurus
PS
paleomag cored
N→R datum (C5AAr base)
FSE
FS
TAM 27 · lower lignite
TAM 29 · pmag base
pol.
FS
Pebas 38 · pmag top / lignite
C5AAr
PS
paleomag cored
FSE
FS (lignite)
Pebas 17 · lignite
Pebas 14 · base
Aligned on the N→R reversal: both sections are Normal below the datum and Reversed at it (shared C5AAn→C5AAr boundary). Lignites are offset → diachronous.
lignite/coal (F3A)
organic clay (F1B)
mudstone (clay) (F1A/B)
silty clay (F2C/D)
sandy clay (F2A/B)
shell bed (F3B)
reversal datum (C5AAr base)
no polarity data (lignite/gap)
Purussaurus
mollusc/shell (F3B)


SC elemental stratigraphy — redox/provenance proxies (Pebas mega-wetland). High Fe–S–Ca = pyritic, reducing, shell-bearing (marine-influenced) intervals.
SC composite scan depth (cm) — young (top) to old (bottom)
Fe (magnetic+authigenic)
S (pyrite)
Ca (carbonate/shell)
Fe/Ti (authigenic Fe)
K/Ti (clay/detrital)
Mn (redox)
pol
N
R
N
N
N


Local (Peru TAM/SC) vs regional (La Venta Honda Gp, U-Pb-calibrated) vs GPTS
Preferred tie: Peru reversal = C5AAr = La Venta Monkey-Beds / Purussaurus interval (~13.2–13.4 Ma)
Age (Ma)
C5An.1n
C5An.2n
C5Ar.1n
C5Ar.2n
C5AAn
C5ABn
C5ACn
palyno T15/T16
floor 12.7 Ma
Monkey Beds /
P. neivensis
(C5AAr–C5Ar.2n)
GPTS
(GTS2020)
La Venta
(Honda Gp)
U-Pb
zircon
SC
(Peru)
TAM
(Peru)



ODP 1237 mid-Miocene sed. rate ~6-9 m/Myr (7.7 at 13 Ma) ≈ our decompacted ~6-9 m/Myr — both condensed.